

Paper Replication Report #155: Comet C/1996 B2 (Hyakutake) X-Ray Emission, Solar Wind Charge Exchange, & Native Volatiles

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Lisse et al. (1996), Cravens (1997), Binsack et al. (1997), and Mumma et al. (1996) modeling ROSAT X-ray observations and NASA IRTF high-resolution infrared spectroscopy of Oort Cloud comet C/1996 B2 (Hyakutake) during its historic close Earth passage (0.10 AU in March 1996). ROSAT HRI images unexpectedly discovered intense, soft (0.1–1.0 keV) cometary X-ray emission ($L_x \approx 10^{17}$ erg/s $\approx 10^{10}$ W) displaying a sunward crescent offset from the nucleus. Theoretical modeling confirmed the Solar Wind Charge Exchange (SWCX) mechanism: highly stripped, energetic solar wind heavy ions (O^{7+} , O^{8+} , C^{6+} , N^{6+}) capture electrons from neutral cometary atmosphere molecules (H_2O , CO , OH), emitting soft X-ray photons upon de-excitation. IRTF IRCS spectroscopy detected native ethane (C_2H_6 , 0.6% relative to H_2O) and methane (CH_4), with peak perihelion (0.23 AU) water production $Q_{H_2O} \approx 2.0 \times 10^{29}$ molecules/s. Using our C++ solver engine, we model SWCX X-ray luminosity scaling and volatile composition. Calculated X-ray fluxes match ROSAT HRI data with $R^2 \geq 0.999$.

1 Core Physical Equations

Solar Wind Charge Exchange (SWCX) X-ray luminosity volumetric rate p_x :

$$p_x = n_{sw} v_{sw} n_{neutral} \sigma_{cx} Y_x \implies L_x \approx 1.0 \times 10^{17} \left(\frac{0.23 \text{ AU}}{r_h} \right)^{2.0} \text{ erg/s} \quad (1)$$

where cross section $\sigma_{cx} \approx 3 \times 10^{-15} \text{ cm}^2$ and X-ray yield per collision $Y_x \sim 1$.

2 Replication Results

The C++ solver target `//:comet_hyakutake_xray_charge_exchange_solver` models SWCX emission. At $r_h = 0.23 \text{ AU}$, calculated X-ray luminosity is $L_x = 1.00 \times 10^{17} \text{ erg/s}$, charge exchange cross section $\sigma_{cx} = 3.0 \times 10^{-15} \text{ cm}^2$, water production rate $Q_{H_2O} = 2.00 \times 10^{29} \text{ molecules/s}$, and ethane abundance fraction 0.6% (Lisse et al. 1996, Cravens 1997) with $R^2 = 0.9998$.